

MUHAMMED SHAMEEL

AI Engineer | Machine Learning Engineer

muhammedshameel3009@gmail.com | +91 8891232625 | Kerala, India

[muhammed-shameel.github.io](https://github.com/muhammed-shameel) | linkedin.com/in/muhammed-shameel

github.com/Muhammed-Shameel

SUMMARY

MSc Data Science and Business Analytics graduate with a Mathematics background and hands-on experience in Machine Learning, Retrieval-Augmented Generation (RAG), and AI-powered application development. Proficient in Python, FastAPI, Scikit-learn, SQL, statistical analysis, and feature engineering, with experience building end-to-end machine learning solutions, data pipelines, and cloud-deployed applications. Passionate about developing scalable AI systems that solve real-world problems.

TECHNICAL EXPERTISE

Programming Language

Python, SAS, SQL

Data Visualization & Tools

Matplotlib, Seaborn, Tableau, SPSS, Jupyter Notebook, VS Code, SAS, SAS Enterprise Minor

Machine Learning & AI

Scikit-learn, TensorFlow, Keras, PyTorch, Random Forest, XGBoost, Support Vector Machines (SVM), Neural Networks, Model Evaluation, Hyperparameter Tuning, Retrieval-Augmented Generation (RAG), Prompt Engineering, RNN, CNN, Deep Learning

Backend & Cloud Computing

FastAPI, API Integration, JSON, Git, GitHub, Docker, Render, Vercel, AWS Fundamentals

Data Engineering

Data Cleaning, Data Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA), Semantic Chunking, Embedding Generation, Metadata Extraction, Vector Databases

Mathematical Foundations

- Linear Algebra
- Probability & Statistics
- Calculus
- Optimization

Statistical Analysis

Hypothesis Testing, Regression Analysis, ANOVA, Correlation Analysis, Factor Analysis, Feature Engineering

WORK EXPERIENCE

Agentic AI Intern

InfoCreon Solutions Private Limited

07/2026 – Present

Kerala, India

Training in the design and implementation of AI agents, focusing on LLM applications, workflow automation, and real-world AI solutions.

Cloud & Data Visualization Engineering Intern

InfoCreon Solutions Private Limited

05/2026 – 07/2026

Kerala, India

- Contributed to the design, development, and deployment of AI-driven software solutions in a collaborative engineering environment.
- Developed backend services using Python and FastAPI, supporting data processing, machine learning integration, and application functionality.
- Applied data preprocessing, feature engineering, statistical analysis, and model evaluation techniques to improve AI-driven solutions.
- Participated in cloud deployment, version control, testing, debugging, and iterative software improvements using modern development practices.
- Worked with cross-functional technologies including Docker, Git, Render, and Vercel while leveraging AI-assisted development tools to improve productivity.

Mathematics Tutor

07/2023 – 03/2024

- Delivered one-on-one mathematics tutoring to undergraduate and high-school students.

- Improved student understanding and academic performance through personalized learning strategies.

PROJECTS

Grocery AI – Retrieval-Augmented Generation (RAG)

07/2026 – Present

- Engineered an advanced Retrieval-Augmented Generation (RAG) system that delivers context-aware responses for grocery storage, food safety, freshness, and shelf-life recommendations using domain-specific knowledge.
- Designed a hybrid retrieval pipeline combining semantic search, metadata filtering, entity extraction, and intent-aware retrieval to improve response relevance and retrieval accuracy.
- Built an automated knowledge processing pipeline incorporating semantic chunking, embedding generation, document parsing, and metadata enrichment for efficient indexing of structured and unstructured data.
- Developed scalable FastAPI services to expose retrieval APIs and support seamless integration with Large Language Models (LLMs).
- Architected the system with a modular design to enable future expansion through automated knowledge ingestion, vector database integration, and multimodal AI capabilities for food recognition and freshness assessment.
- Tech Stack: Python, FastAPI, Sentence Transformers, RAG, Hybrid Search, Semantic Chunking, Vector Embeddings, LLMs, Document Processing, Metadata Engineering

FLUVIO-AI-Powered Flood Intelligence Dashboard

05/2026 – 07/2026

- Developed an AI-powered flood intelligence dashboard integrating machine learning predictions with interactive geospatial visualization and real-time analytics.
- Built a FastAPI backend for machine learning inference, data processing, and analytics APIs powering the dashboard.
- Performed data preprocessing and feature engineering to improve flood risk prediction using environmental and geospatial data.
- Designed an interactive analytics interface with region-based filtering, KPI reporting, historical and live data visualization, and export functionality.
- Deployed the backend on Render and frontend on Vercel using Docker and Git-based deployment workflows.
- Tech Stack: Python, FastAPI, Scikit-learn, Next.js, TypeScript, Docker, Render, Vercel, Git

A Data-Driven Analysis on Football Players Under Different Managers

- Analyzed 2,500+ match records spanning multiple Premier League seasons.
- Applied Factor Analysis and feature engineering to derive latent player performance metrics.
- Evaluated multiple machine learning models, achieving 68.7% accuracy and 0.65 macro F1-score with XGBoost.
- Generated tactical insights into manager-player compatibility and performance adaptation.
- Tech Stack: Python (Pandas, NumPy, Scikit-learn), SQL, Matplotlib, Seaborn, Flask

CSIRO Image2Biomass Prediction

- Developed regression models using MSE, Huber, and LogCosh loss functions.
- Achieved $R^2 = 0.86$ through ensemble learning and hyperparameter optimization.
- Built a weighted ensemble model to improve prediction stability across heterogeneous data.
- Tech Stack: Python, TensorFlow, Keras, NumPy, Matplotlib

EDUCATION

Master of Science in Data Science and Business Analytics

03/2026

Asia Pacific University of Technology & Innovation,
Malaysia
GPA: 3.62

Bachelor of Science in Mathematics

04/2024

Kottakkal Farook Arts and Science College (Affiliated
under University of Calicut)